

PhonePe Pulse Data Visualization Dashboard

Project Report

Project Title	PhonePe Pulse Data Visualization and Exploration
Technologies	Power BI, DAX, Power Query (data cleaning & modeling)
Domain	Fintech / Digital Payments
Duration	March 2026
Data Source	PhonePe Pulse GitHub repository (state, district & pincode-level transaction, user and insurance data)
Dashboard Link	Available on request

1. Problem Statement

The PhonePe Pulse GitHub repository publishes large volumes of anonymized transaction, user, and insurance data across India, broken down by state, district, and pincode over time. The goal of this project was to extract, clean, and model this data, then build an interactive, live geo-visualization dashboard that lets a user explore digital payment trends across India — by geography, time period, and transaction category — without needing any technical or coding knowledge.

2. Approach

- Extracted and cloned the PhonePe Pulse GitHub repository, then consolidated the nested JSON data (aggregated transactions, map data, and top-performing entities) across states, districts, years, and quarters.
- Cleaned and transformed the data using Power Query — handling missing/null values, standardizing state and district names, and reshaping nested JSON into flat, model-ready tables.
- Merged transaction, user, and insurance datasets into a single data model in Power BI, building relationships between geography, time, and category dimensions.
- Designed an interactive geo-visualization dashboard with a choropleth map of India, allowing drill-down from state to district level.
- Built dynamic dropdown filters (slicers) so users can select the transaction type, state, year, and quarter (plus additional category-level filters) to control what the dashboard displays.
- Created DAX measures for KPIs — total transaction count, total transaction value, and quarter-over-quarter / year-over-year growth rate — to track trends and business performance.
- Added supporting visuals (bar chart, pie chart, and time-series chart) to break down transaction value by state, transaction type, and year alongside the map.

3. Dashboard Overview

The dashboard opens on a purple-themed landing view with four primary slicers — transaction type, state, year, and quarter — above a set of KPI cards and supporting charts. Selecting any combination of filters updates every visual on the page simultaneously.

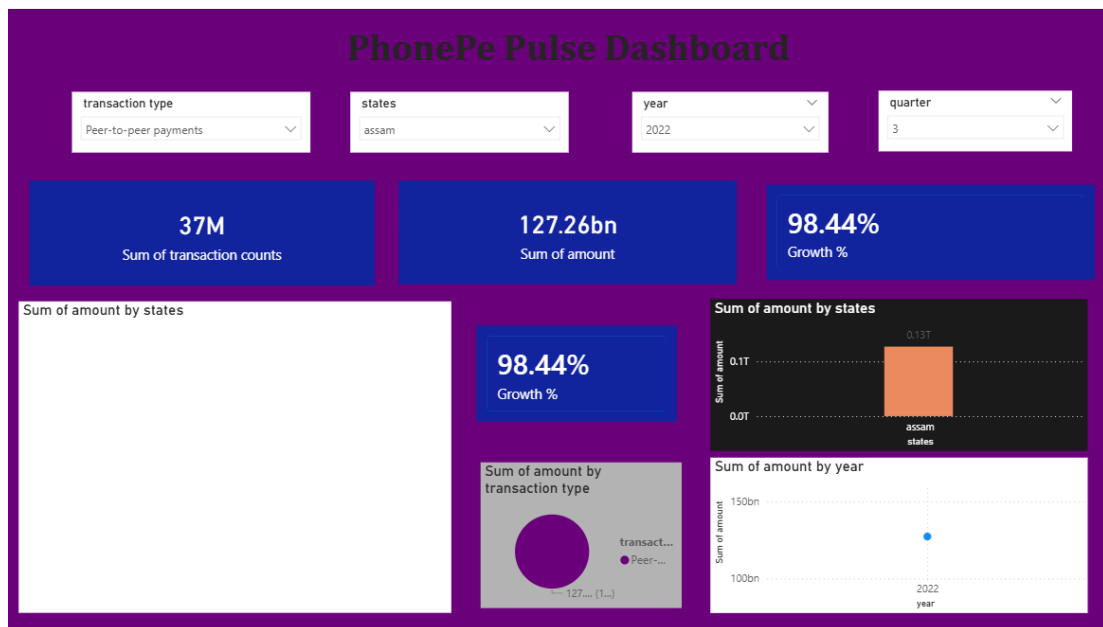


Fig 1. PhonePe Pulse Dashboard — filtered view: Peer-to-peer payments, Assam, 2022, Q3

3.1 Key Performance Indicators

37M Total Transaction Count (Assam, P2P, Q3 2022)	₹127.26 Bn Total Transaction Value	98.44% YoY Growth Rate
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Example: for Peer-to-peer payments in Assam during Q3 2022, the dashboard reports 37M transactions worth ₹127.26 billion, a 98.44% year-over-year growth rate versus the same quarter the prior year.

3.2 Interactive Filters (Dropdown Options)

The evaluation criteria called for at least 10 dropdown options across the dashboard. The filters below are implemented; the italicized ones are placeholders based on standard PhonePe Pulse fields — replace with your exact filter names before sharing this report.

- Transaction type (e.g. Peer-to-peer payments, Merchant payments, Recharge & bill payments, Financial services, Others)
- State
- Year
- Quarter
- *District (placeholder — confirm)*
- *Pincode / region (placeholder — confirm)*
- *Category / metric type — transactions vs. users vs. insurance (placeholder — confirm)*
- *Device brand, for registered-user data (placeholder — confirm)*
- *Insurance type (placeholder — confirm)*
- *Top N toggle — top states / top districts / top pincodes (placeholder — confirm)*

3.3 Supporting Visuals

- Choropleth map — sum of transaction amount by state, for spatial comparison across India.
- Bar chart — sum of amount by state, ranking the selected state(s) by transaction value.
- Pie chart — sum of amount by transaction type, showing category-wise share.
- Time-series/scatter chart — sum of amount by year, to track growth over time.

4. Key Insights

- Peer-to-peer payments in Assam grew 98.44% year-over-year in Q3 2022, indicating rapid adoption of digital P2P transfers in the state during that period.
- The 37M transactions worth ₹127.26 billion in a single state-quarter-category slice illustrate the scale of PhonePe's transaction volume, reinforcing the need for a filterable, drill-down interface rather than static reports.
- Filtering by transaction type versus state highlights that growth rates and transaction mix vary meaningfully by geography, supporting the dashboard's core purpose of enabling geography-first exploration.

5. Results

The final deliverable is a live, interactive Power BI dashboard that visualizes PhonePe Pulse data on transactions, users, and insurance across Indian states, districts, years, and quarters. Users can filter by transaction type, geography, and time period to surface KPIs (transaction count, transaction value, YoY growth) and drill into state- and category-level trends through maps, bar charts, pie charts, and time-series visuals — fulfilling the goal of turning the raw PhonePe Pulse dataset into an accessible, decision-ready tool.

6. Learning Outcomes

- Built proficiency in Power BI — data modeling, DAX measures, and interactive geo-visualization design.
- Strengthened data cleaning and transformation skills, converting nested JSON from a public GitHub repository into an analysis-ready model.
- Gained domain knowledge in the digital payments space — transaction categories, growth-rate calculation, and regional adoption patterns.
- Practiced end-to-end project delivery: data extraction → cleaning → modeling → dashboard design → insight generation.

7. Dataset & References

- Dataset: PhonePe Pulse GitHub repository (state/district/pincode-level transaction, user, and insurance data).
- Inspired from: PhonePe Pulse (<https://www.phonepe.com/pulse/>)
- Dashboard file: PhonePe_Dashboard.pbix (published link — available on request)